Examiner only

3. *(a)* (i) State Kepler's 1st and 2nd laws. [3]

(ii) Kepler's 3rd law can be derived from Newton's gravitational law and the equation for centripetal motion. Show that, for any object in a circular orbit about the Earth:

$$T^2 = \frac{4\pi^2}{GM_E} r^3$$

where T = the period of orbit, r = the radius of orbit, G = the gravitational constant and M_E = the mass of the Earth. [3]

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(b) The radius of the geostationary orbit above the Earth's equator is 42 000 km and the Earth-Moon distance is 380 000 km. Use Kepler's 3rd law to calculate the period of the Moon's orbit. [3]

(c) Estimate the minimum possible orbital period for a satellite orbiting the Earth, stating any assumption that you make (the radius of the Earth is 6 370 km). [3]

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